PROJECT PREP

STATISTICS DECISION TREE

Find the row that matches your experiment, then use the method next to it. Start at the top.

START: WHAT ARE YOU COMPARING?

One group before and after a change
Compare the two averages and describe the difference. Show it as a bar chart with two bars.

Example: heart rate before and after music

Two or more separate groups
Calculate the average and the range for each group. Compare the averages, then check whether the ranges overlap. Heavy overlap means your result is less certain.
Example: three paper towel brands

One thing measured repeatedly over time
Plot a line graph with time on the horizontal axis. Describe the trend in words: rising, falling, leveling off.

Example: plant height each day for two weeks

Two numbers measured together on the same subject
Plot a scatter plot. Describe whether higher values of one go with higher values of the other.

Example: sleep hours against reaction time

THREE NUMBERS THAT COVER MOST MIDDLE SCHOOL PROJECTS

Mean, the average	Add every trial and divide by the number of trials. This is your headline number.
Range	The largest value minus the smallest. A wide range means your results varied a lot.
Number of trials	Report it every time. Three trials is the minimum, and more is better.

YOU DO NOT NEED ADVANCED STATISTICS

Middle school judges look for careful measurement, enough trials, and an honest reading of the data. A clear average with a stated range beats a test you cannot explain.